
IOT-BASED SINGLE-AXIS SOLAR TRACKER WITH BATTERY MONITORING AND CLOUD DASHBOARD

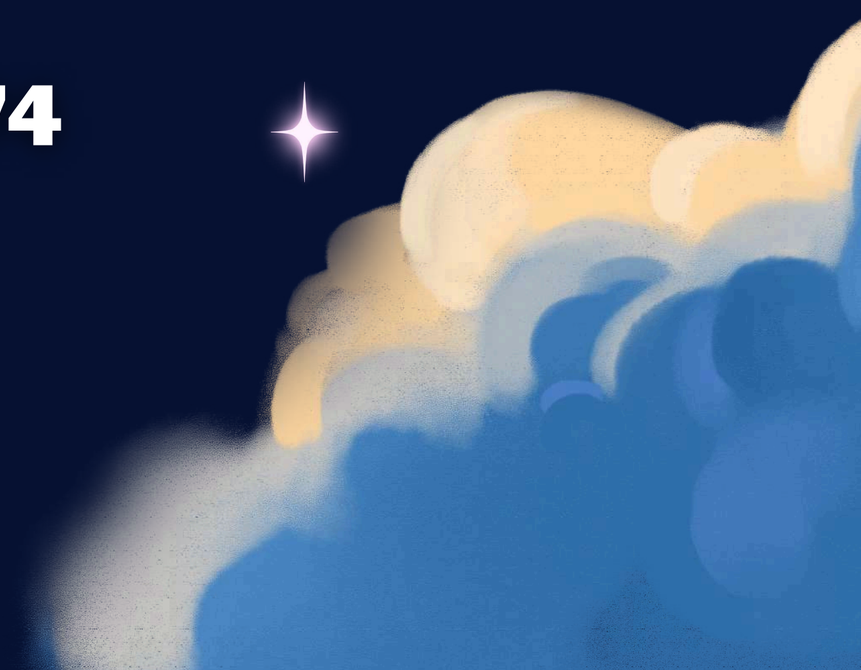
MEMBER

HOÀNG TIẾN MẠNH-SE203727

VÕ ĐẶNG ANH KHOA-SE203865

ĐẶNG THỊ PHƯƠNG ANH-SE203874

NGUYỄN THÀNH LUÂN-SE184179



1. INTRODUCTION

Fixed panel

- Peaks only around solar noon; cosine loss grows quickly in the morning and afternoon.

Sun tracking

- Studies report roughly 20–40% more energy than a fixed panel.

Why single-axis

- One actuator, one hinge and simple control; a good fit for low-power prototypes.

Sensor-driven closed loop: no GPS, real-time clock or site calibration needed.

2. METHOD AND MATERIALS

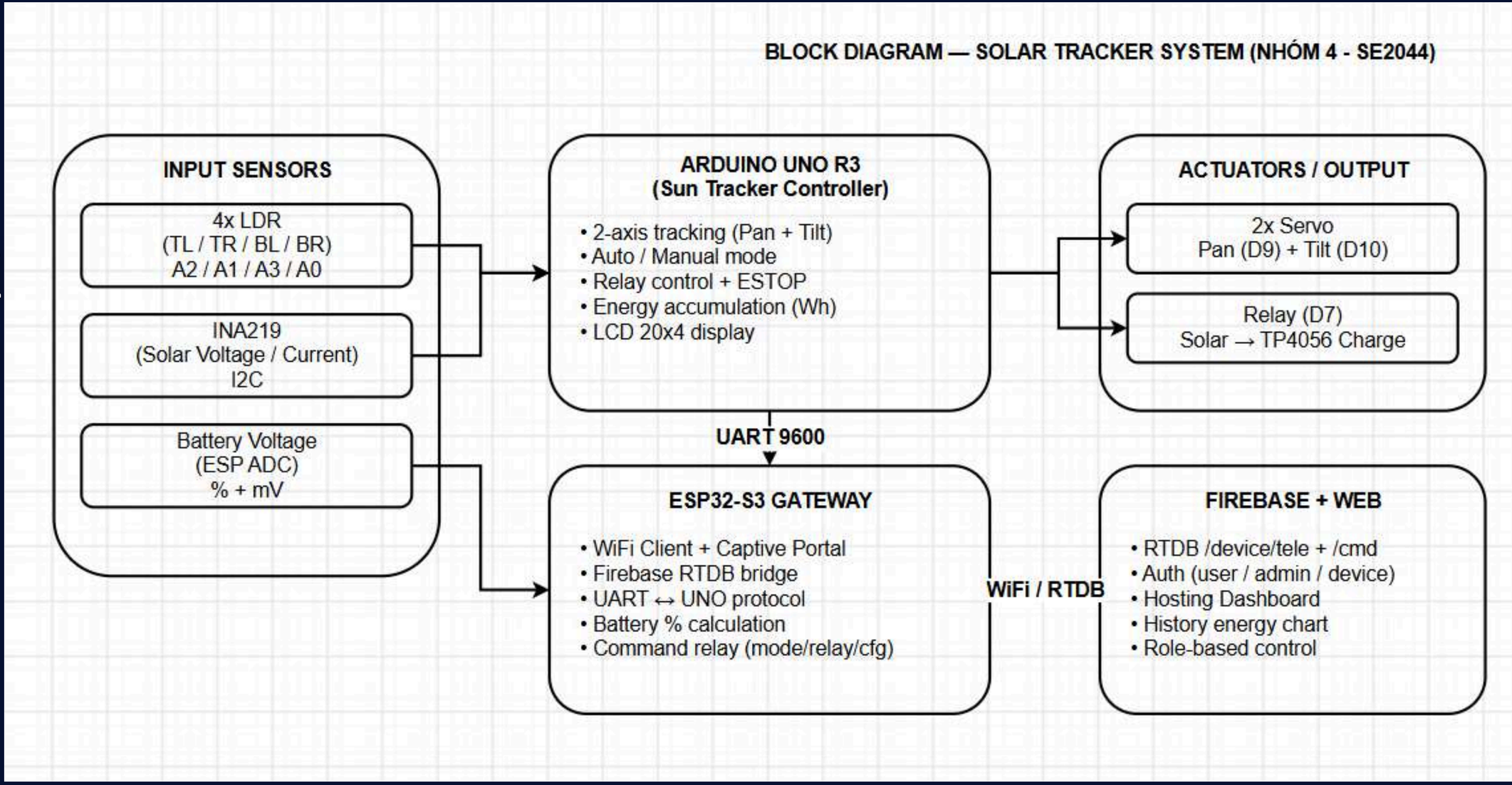
- Microcontroller: ESP32-S3 DevKit N16R8 (integrated Wi-Fi, dual-core processor, 12-bit ADC).
- Light Sensors: Four photoresistor (LDR) sensor modules mounted at the four corners to measure light intensity differences.

Power System:

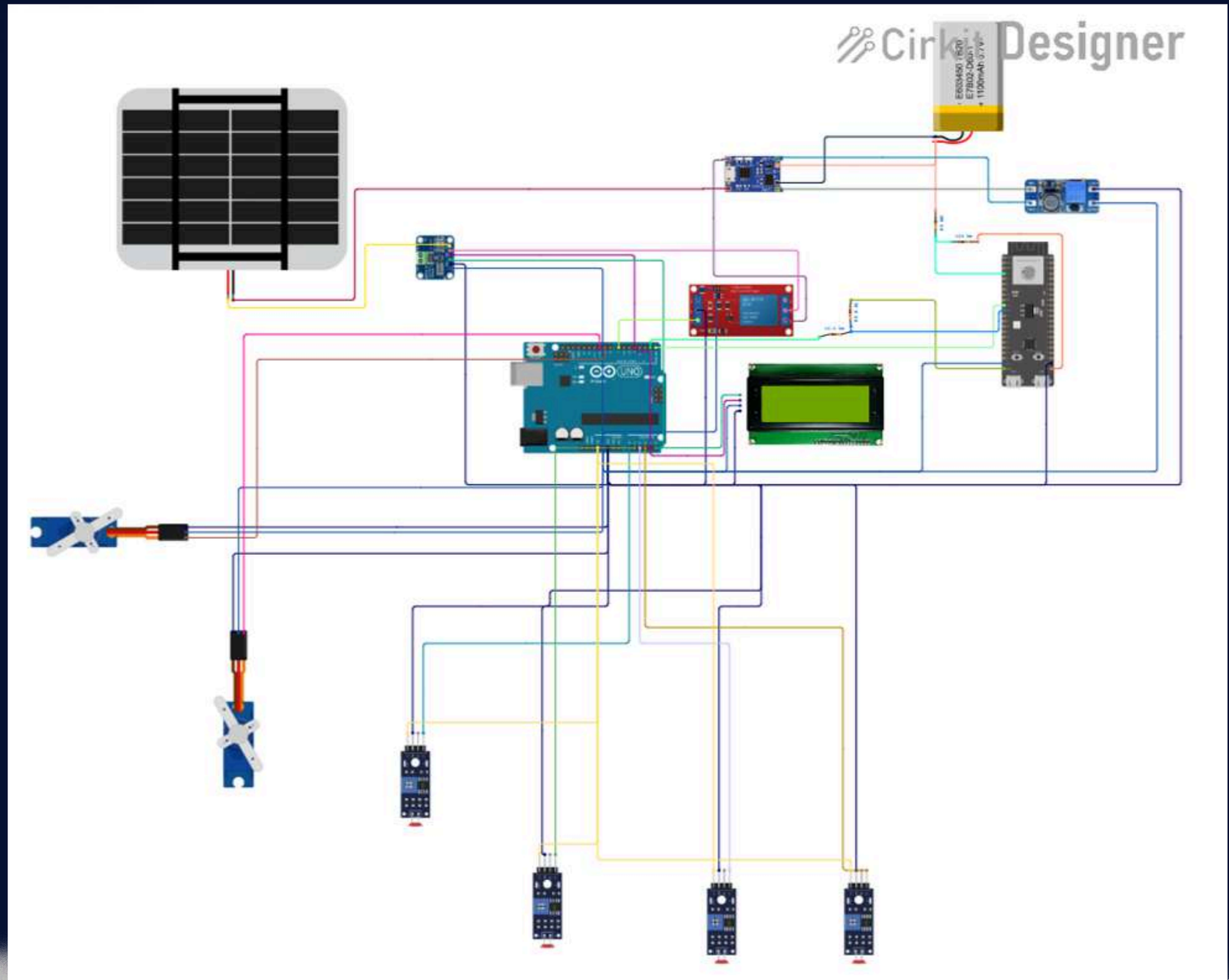
- 5W solar panel.
- TP4056 charging module (with battery protection).
- One 18650 Li-ion cell (2500 mAh).
- MT3608 buck converter adjusted to 5V to exclusively supply the servo motor, preventing motor current from causing voltage drops in the microcontroller.
- Micro Servo SG90: Rotation solar panel to where has most lightest .



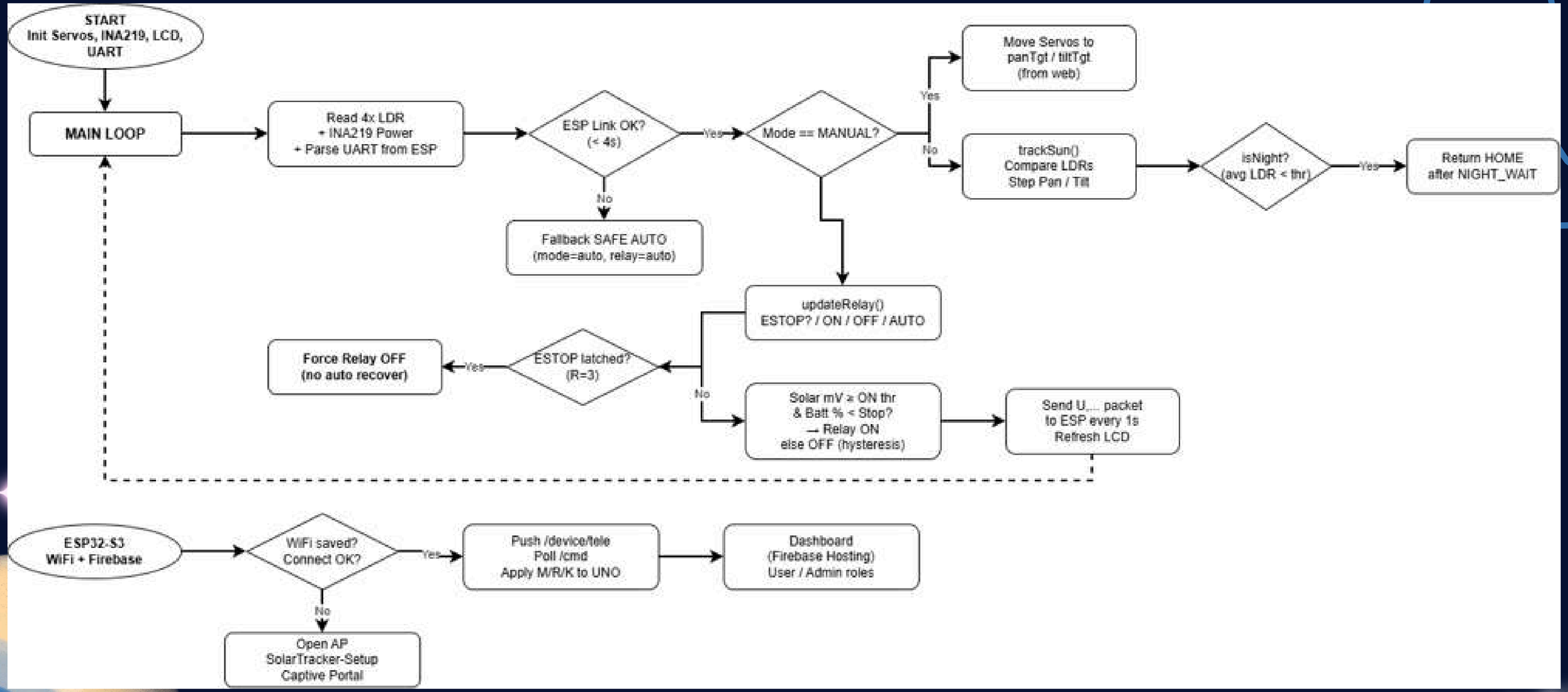
2.1 BLOCK DIAGRAM



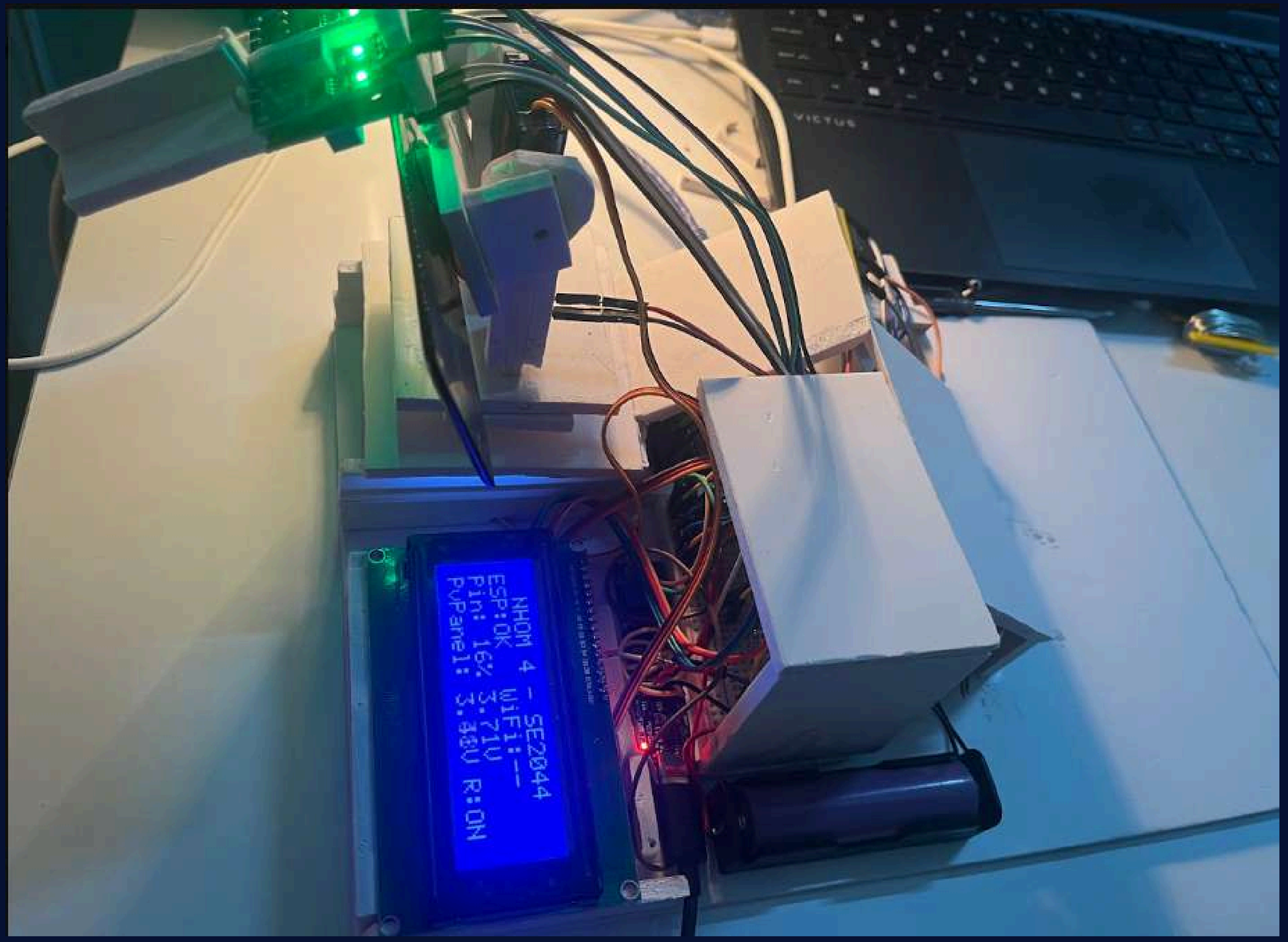
3. ELECTRONIC CIRCUIT

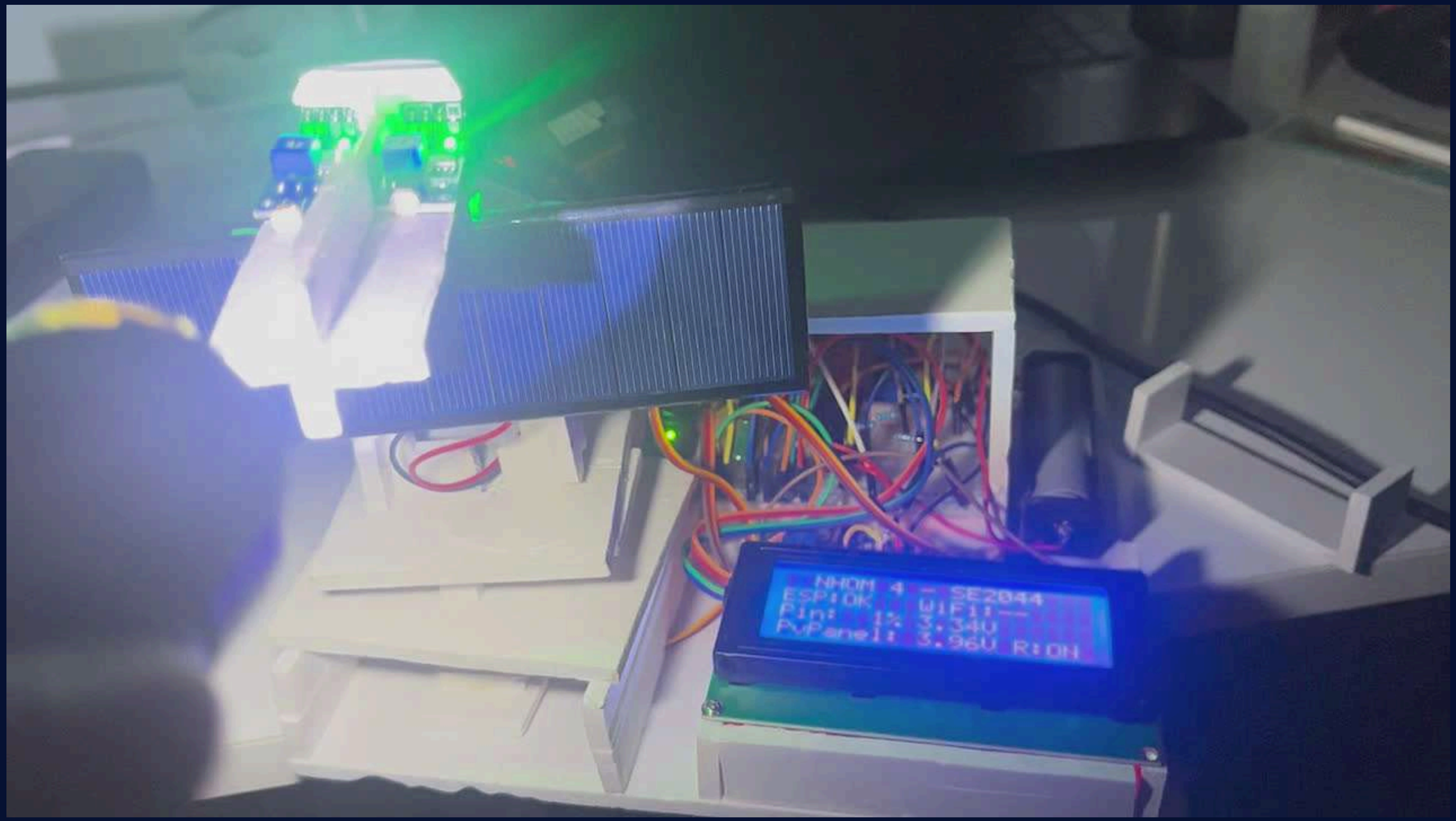


4. FLOWCHART


START
Init Servos, INA219, LCD, UART
MAIN LOOP
Read 4x LDR
+ INA219 Power
+ Parse UART from ESP
ESP Link OK?
(< 4s)
Yes
No
Fallback SAFE AUTO
(mode=auto, relay=auto)
Mode == MANUAL?
Yes
No
Move Servos to
panTgt / tiltTgt
(from web)
trackSun()
Compare LDRs
Step Pan / Tilt
isNight?
(avg LDR < thr)
Yes
Return HOME
after NIGHT_WAIT
updateRelay()
ESTOP? / ON / OFF / AUTO
ESTOP latched?
(R=3)
Yes
Force Relay OFF
(no auto recover)
No
Solar mV ≥ ON thr
& Batt % < Stop?
→ Relay ON
else OFF (hysteresis)
Send U,... packet
to ESP every 1s
Refresh LCD
ESP32-S3
WiFi + Firebase
WiFi saved?
Connect OK?
Yes
No
Open AP
SolarTracker-Setup
Captive Portal
Push /device/tele
Poll /cmd
Apply M/R/K to UNO
Dashboard
(Firebase Hosting)
User / Admin roles

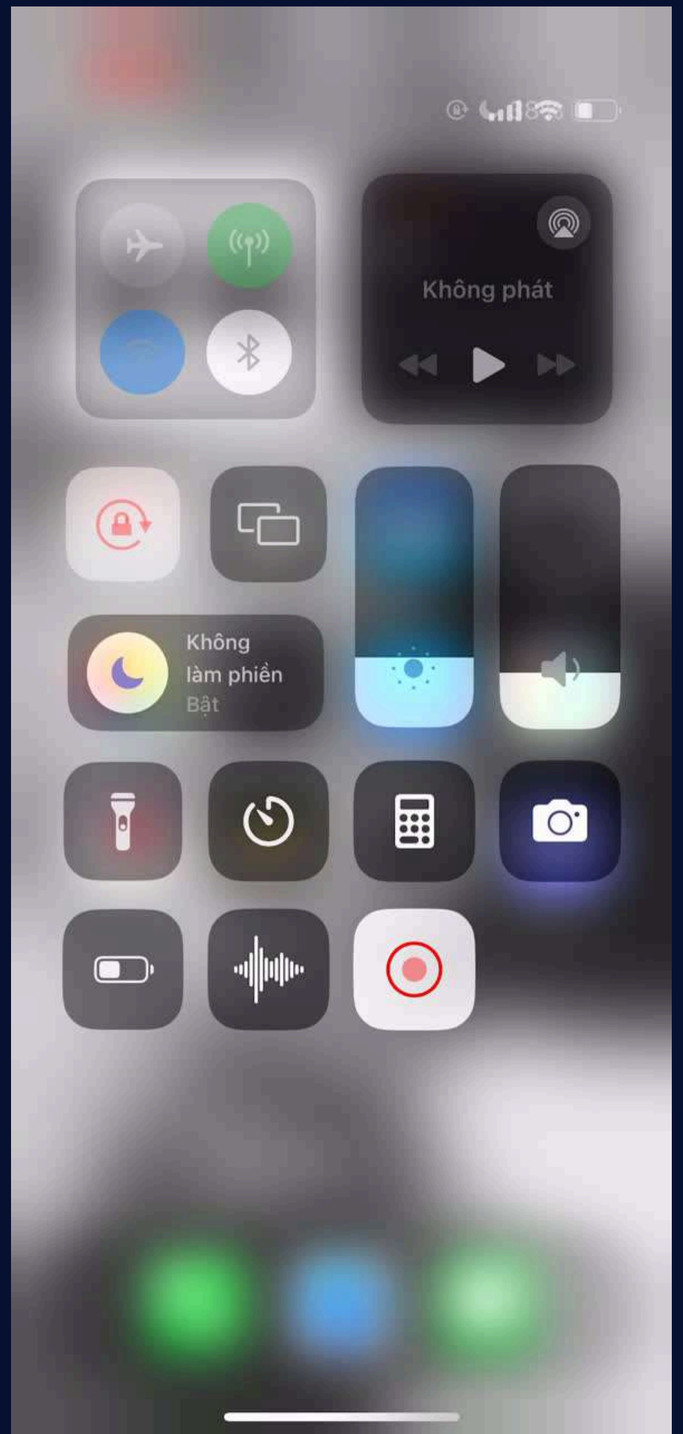
LẮP MẠCH





VIDEO DEMO

CONFIG WIFI FOR SYSTEM CONNECT INTERNET



5. RESULTS AND CONCLUSION

- Prototype Testing: The system operates stably with smooth rotation, avoiding continuous oscillation thanks to the implemented deadband hysteresis.

- The project has successfully built a low-cost single-axis solar tracking system that optimizes solar energy harvesting, integrates comprehensive hardware/software safety protection mechanisms, and provides online IoT monitoring.

THANK YOU!

